

BRAGANCA, Portugal

WMO: 085750

Lat: 41.8037N

Lon: 6.7429W

Elev: 692

StdP: 93.29

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	-2.5	-8.1	2.1	2.7	-6.2	2.4	2.3	9.7	7.8	8.3	7.8	0.6	340	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	33.8	18.4	32.1	18.1	30.5	17.5	19.7	30.4	19.0	29.6	18.3	28.3	2.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.7	21.9	15.5	12.0	21.2	14.7	11.4	20.8	59.1	30.3	56.5	29.6	54.2	28.4	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.4	5.4	-6.2	36.3	1.5	1.0	-7.3	37.1	-8.2	37.6	-9.1	38.2	-10.2	39.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-6.7	21.6	1.6	2.6	-7.8	23.5	-8.8	25.0	-9.7	26.5	-10.8	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	5.1	6.4	9.3	11.1	14.5	19.0	21.7	21.6	18.2	13.7	8.2	5.5
	DBStd	6.77	3.04	2.83	3.18	3.59	3.77	3.80	3.31	3.19	3.21	3.14	3.09	3.02
	HDD10.0	562	153	104	52	29	5	0	0	0	6	69	143	
	HDD18.3	2316	410	336	280	218	128	37	9	8	42	146	303	399
	CDD10.0	1622	2	3	31	62	145	271	363	361	245	121	17	2
	CDD18.3	332	0	0	0	1	10	58	114	110	37	3	0	0
	CDH23.3	4146	0	0	1	25	157	753	1379	1326	451	53	0	0
	CDH26.7	1639	0	0	0	1	34	283	602	578	133	8	0	0
Wind	WSAvg	2.1	1.9	2.2	2.4	2.4	2.3	2.2	2.2	2.1	1.8	1.7	1.9	1.7
Precipitation	PrecAvg	739	104	83	64	62	60	36	18	15	42	87	89	97
	PrecMax	1152	383	299	328	170	141	117	121	86	150	248	244	349
	PrecMin	467	0	0	0	1	3	4	0	0	0	0	0	3
	PrecStd	181	79	68	63	41	35	28	22	18	37	60	66	84
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	18.1	22.4	25.9	30.0	34.4	35.5	36.3	33.1	27.7	18.7	14.1
		MCWB	9.5	10.3	11.5	14.3	17.2	18.7	19.1	18.8	17.5	14.8	12.9	10.8
	2%	DB	12.4	15.6	20.3	23.6	27.1	31.8	33.7	33.7	30.0	24.6	16.3	12.6
		MCWB	9.5	9.5	10.7	13.3	16.1	18.2	18.4	18.4	16.8	14.7	11.7	10.4
	5%	DB	11.2	13.6	18.4	21.4	24.9	29.9	32.1	31.9	27.9	22.0	14.6	11.4
		MCWB	9.1	8.8	10.3	12.5	15.1	17.5	18.2	17.8	15.9	14.2	11.3	9.4
	10%	DB	10.0	11.7	16.3	18.8	22.6	28.0	30.2	30.1	26.0	19.8	13.2	10.2
		MCWB	8.2	8.1	9.7	11.6	14.2	16.7	17.7	17.3	15.6	13.6	10.6	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	12.0	13.0	15.7	18.6	20.2	21.6	20.9	18.7	17.5	14.7	12.2
		MCDB	13.0	15.4	18.2	23.9	27.9	30.2	31.4	31.9	28.4	22.3	17.0	13.0
	2%	WB	10.5	10.7	12.0	14.3	17.0	19.2	19.9	19.4	17.7	16.3	13.1	11.0
		MCDB	11.6	13.8	17.2	21.9	24.9	29.3	30.4	30.9	27.2	20.9	14.8	12.3
	5%	WB	9.5	9.6	11.1	13.0	15.8	18.4	19.0	18.6	17.0	15.4	12.0	9.8
		MCDB	10.7	12.4	16.4	20.0	23.5	28.1	29.5	29.5	25.8	19.8	13.7	10.9
	10%	WB	8.4	8.6	10.3	11.9	14.8	17.5	18.2	17.9	16.2	14.6	11.1	8.8
		MCDB	9.7	11.3	15.1	17.6	21.4	26.5	28.4	27.9	23.7	18.4	12.9	9.9
Mean Daily Temperature Range	5% DB	MDBR	7.5	9.7	10.9	10.9	11.8	13.5	14.5	14.9	13.1	10.4	8.3	7.5
		MCDBR	9.6	13.5	15.3	15.4	15.5	16.2	16.5	17.1	16.4	14.9	10.7	8.7
		MCWBR	6.8	8.3	8.1	7.3	7.0	6.5	6.1	6.1	6.5	7.1	6.8	6.2
	5% WB	MCDBR	6.8	10.9	12.7	14.0	14.1	15.1	15.5	15.7	14.1	11.0	7.9	6.4
		MCWBR	5.3	7.4	7.2	7.2	6.6	6.4	6.3	6.0	6.0	5.9	5.7	5.1
Clear-Sky Solar Irradiance	taub	0.291	0.302	0.333	0.341	0.351	0.353	0.339	0.348	0.336	0.330	0.300	0.288	
	taud	2.541	2.504	2.437	2.422	2.418	2.423	2.442	2.410	2.463	2.498	2.542	2.546	
	Ebn at Noon	871	913	916	929	923	918	927	910	900	865	850	844	
	Edh at Noon	70	84	102	111	114	114	111	111	98	84	69	64	
All-Sky Solar Radiation	RadAvg	1.76	2.78	3.92	5.04	6.22	7.15	7.45	6.59	5.02	3.15	1.99	1.56	
	RadStd	0.25	0.39	0.58	0.62	0.63	0.41	0.31	0.35	0.31	0.38	0.35	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+34

Nomenclature: See separate page